

WOENS DRECHT, Netherlands

WMO: 063400

Lat: 51.4478N

Lon: 4.3419E

Elev: 19

StdP: 101.09

Time Zone: 1.00 (EUC)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.2	-4.8	-10.2	1.6	-5.5	-7.8	1.9	-3.9	12.8	9.6	10.9	8.9	2.6	70	0.548

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.5	29.3	19.9	26.9	19.0	25.0	18.1	21.0	26.9	20.0	25.2	19.1	23.6	3.5	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.1	13.9	23.0	18.1	13.1	21.9	17.3	12.4	21.1	60.7	26.9	57.2	25.3	54.2	23.7	25.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.8	8.4	7.5	-10.7	33.2	2.9	1.7	-12.8	34.4	-14.5	35.4	-16.1	36.3	-18.2	37.5
			-11.0	23.1	2.9	0.7	-13.1	23.6	-14.8	24.0	-16.4	24.4	-18.5	24.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.5	3.4	4.0	6.4	9.7	13.1	16.0	18.0	17.8	15.0	11.4	7.1	4.2
	DBStd	6.31	4.43	4.02	3.51	3.68	3.51	3.13	2.97	2.76	2.78	3.46	3.58	4.20
	HDD10.0	863	205	169	120	50	11	0	0	0	1	26	99	183
	HDD18.3	2928	462	401	371	261	167	85	43	43	106	216	337	438
	CDD10.0	1059	1	2	7	40	106	179	248	243	151	69	12	3
	CDD18.3	83	0	0	0	1	3	14	32	28	6	1	0	0
	CDH23.3	814	0	0	0	13	57	158	290	250	44	2	0	0
	CDH26.7	220	0	0	0	1	8	42	87	75	7	0	0	0
Wind	WSAvg	3.5	4.0	3.9	3.8	3.4	3.5	3.3	3.3	3.0	3.0	3.3	3.3	3.8
Precipitation	PrecAvg	845	69	62	49	44	60	74	86	83	75	72	85	86
	PrecMax	1120	141	128	108	97	106	123	177	218	225	170	131	179
	PrecMin	612	4	18	8	0	18	29	21	14	13	24	9	25
	PrecStd	113	33	26	27	24	26	31	39	49	51	30	31	43
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.0	14.6	18.8	25.1	27.9	30.8	32.4	31.6	27.4	22.8	17.0	13.7
		MCWB	11.1	9.6	12.2	15.0	18.8	20.1	21.1	20.9	19.2	17.4	14.2	11.8
	2%	DB	11.6	12.2	16.0	21.8	25.0	27.4	29.1	28.9	24.2	19.7	14.9	12.5
		MCWB	10.1	9.7	11.2	13.7	16.3	19.0	19.8	20.0	18.3	16.1	12.8	11.0
	5%	DB	10.4	11.0	13.7	18.9	22.6	24.8	26.5	26.1	22.0	17.9	13.5	11.3
		MCWB	9.0	9.0	10.1	12.5	15.6	17.9	19.1	19.2	17.0	15.0	11.7	9.9
	10%	DB	9.2	9.7	12.0	16.2	20.1	22.5	24.2	23.6	20.0	16.5	12.2	10.0
		MCWB	8.0	8.2	9.3	11.4	14.7	17.0	18.3	18.2	16.0	14.2	10.7	8.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.6	11.2	13.2	16.2	19.7	21.7	22.2	22.3	20.3	17.9	14.6	12.4
		MCDB	12.5	13.0	16.8	23.1	25.6	27.9	28.8	28.6	25.4	21.4	16.0	13.2
	2%	WB	10.3	10.2	11.8	14.3	17.7	19.9	21.0	20.9	18.9	16.5	13.0	11.2
		MCDB	11.4	11.5	14.8	19.7	22.5	25.2	27.0	26.7	23.0	19.0	14.5	12.3
	5%	WB	9.1	9.3	10.7	13.0	16.3	18.7	20.0	19.7	17.7	15.5	12.0	10.1
		MCDB	10.2	10.7	13.0	17.8	20.9	23.8	25.2	24.5	20.9	17.5	13.4	11.2
	10%	WB	8.0	8.2	9.7	11.9	15.1	17.5	18.9	18.8	16.8	14.5	10.8	8.8
		MCDB	9.1	9.7	11.7	15.6	19.4	21.4	23.1	22.8	19.4	16.2	12.0	9.9
Mean Daily Temperature Range	5% DB	MDBR	5.2	6.2	8.0	10.1	9.9	9.9	9.5	9.7	9.1	7.6	5.8	4.8
		MCDBR	5.1	7.4	11.3	14.4	14.5	14.2	13.9	14.0	11.6	8.7	6.1	4.6
		MCWBR	4.6	5.4	7.3	7.4	7.1	7.1	6.2	6.4	6.3	5.6	4.7	4.0
	5% WB	MCDBR	5.0	6.1	9.5	12.4	11.9	12.4	11.9	12.1	10.1	8.2	5.9	4.6
		MCWBR	4.7	5.0	6.6	6.9	6.7	6.8	6.0	6.2	5.9	5.6	4.9	4.3
Clear-Sky Solar Irradiance	taub	0.335	0.357	0.391	0.422	0.431	0.427	0.433	0.423	0.398	0.372	0.349	0.333	
	taud	2.354	2.330	2.239	2.186	2.208	2.242	2.230	2.275	2.336	2.389	2.369	2.331	
	Ebn at Noon	666	755	794	813	824	831	817	805	782	729	642	599	
	Edh at Noon	64	85	111	132	135	132	131	119	101	80	63	57	
All-Sky Solar Radiation	RadAvg	0.75	1.43	2.61	4.21	5.03	5.44	5.11	4.36	3.22	1.88	0.88	0.57	
	RadStd	0.09	0.28	0.36	0.44	0.41	0.43	0.60	0.38	0.28	0.15	0.09	0.07	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (77 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page